

Hospital Management System

Entity Relationship Diagram (ERD)

A relational database design modeling the core entities, relationships, and operational workflows of a hospital information system.

9 Entities

3NF Normalized

Advanced Modelling

DEPARTMENT		
int	Department_ID	PK
string	Department_Name	
int	Head_Doctor_ID	

Person (supertype)		
int	Person_ID	PK
string	full_name	
date	Date_of_Birth	
string	Gender	
string	Phone	
string	Address	
string	Person_Type	Discriminator → 'Doctor' / 'Patient'

ROOM		
int	Room_ID	PK
string	Room_Number	
string	Room_Type	
string	Status	

Patient(Subtype)		
int	Patient_ID	PK
int	Doctor_ID	FK
int	Room_ID	FK

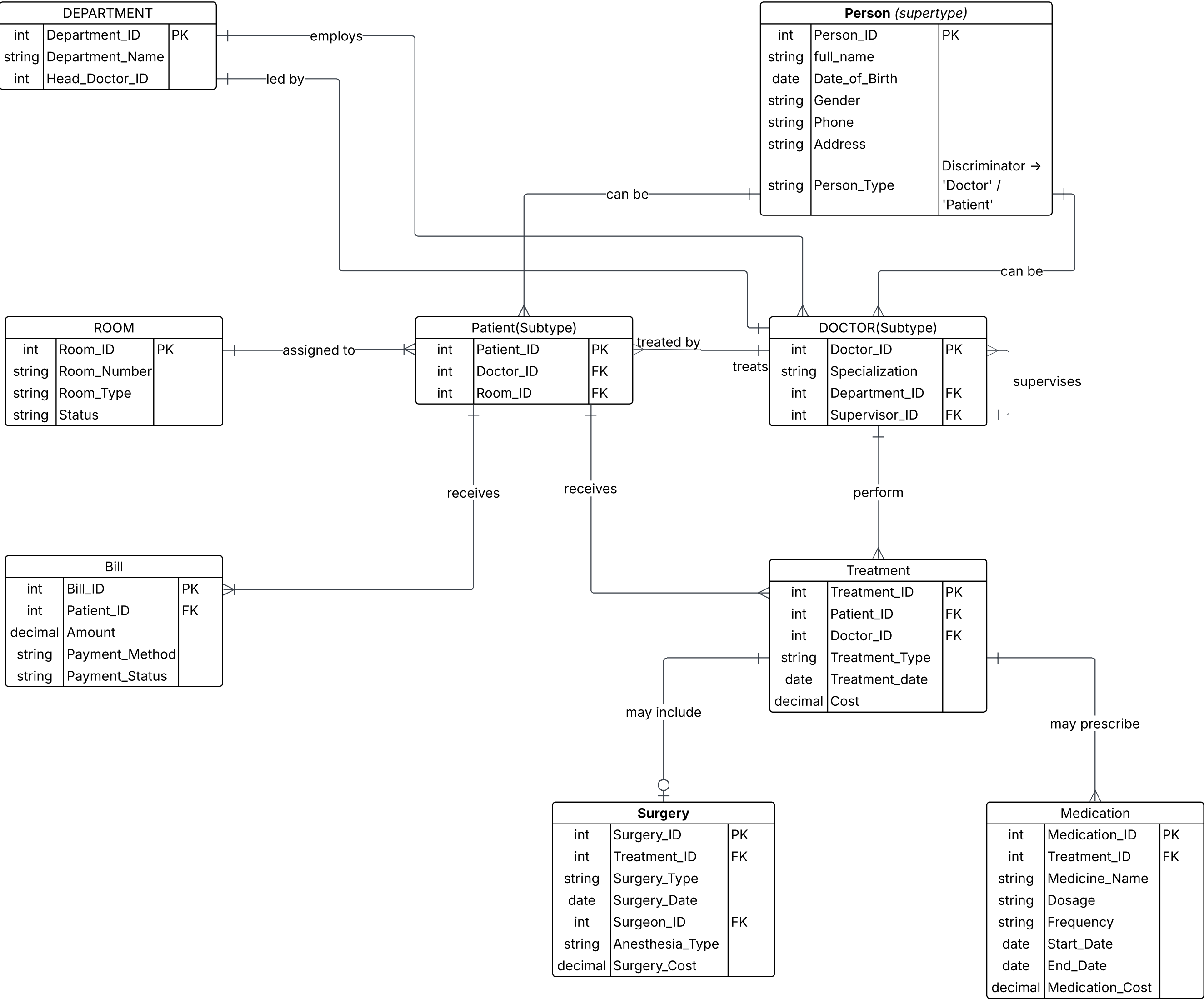
DOCTOR(Subtype)		
int	Doctor_ID	PK
string	Specialization	
int	Department_ID	FK
int	Supervisor_ID	FK

Bill		
int	Bill_ID	PK
int	Patient_ID	FK
decimal	Amount	
string	Payment_Method	
string	Payment_Status	

Treatment		
int	Treatment_ID	PK
int	Patient_ID	FK
int	Doctor_ID	FK
string	Treatment_Type	
date	Treatment_date	
decimal	Cost	

Surgery		
int	Surgery_ID	PK
int	Treatment_ID	FK
string	Surgery_Type	
date	Surgery_Date	
int	Surgeon_ID	FK
string	Anesthesia_Type	
decimal	Surgery_Cost	

Medication		
int	Medication_ID	PK
int	Treatment_ID	FK
string	Medicine_Name	
string	Dosage	
string	Frequency	
date	Start_Date	
date	End_Date	
decimal	Medication_Cost	



Entities & Relationships

What the database models and how the components connect



People

Person (supertype) with Patient & Doctor subtypes — sharing demographics via inheritance, with a discriminator to distinguish roles.



Hospital Resources

Department, Room, and Bill — capturing org structure, room availability, and patient billing with payment tracking.



Clinical Actions

Treatment, Surgery, and Medication — linked via an exclusive arc ensuring each treatment maps to exactly one clinical type.

Key Relationships

Department → Doctor

1 – M

A department employs many doctors; each doctor belongs to one department.

Department → Head Doctor

1 – 1

Each department has exactly one head doctor.

Doctor → Patient

1 – M

One doctor treats many patients across their stay.

Patient → Room

Non-transferable

Room assignment locked until formally updated — ensures data integrity.

Treatment → Surgery / Medication

Arc (exclusive)

A treatment leads to surgery or medication — never both at once.

Doctor → Doctor

Recursive

Senior doctors supervise junior doctors within the same entity.

Design Highlights

Advanced modelling techniques applied in this ERD



Supertype / Subtype

Person is the supertype; Patient and Doctor are subtypes inheriting shared attributes. A discriminator (Person_Type) identifies each role.



Arc Relationship

Treatment uses an exclusive-or arc — it must result in either a Surgery or a Medication record, enforcing mutual exclusivity at the model level.



Recursive Relationship

The Doctor entity references itself via Supervisor_ID, modeling a hierarchy where senior surgeons can supervise junior colleagues.



Third Normal Form (3NF)

All attributes are atomic with no partial or transitive dependencies, ensuring data integrity, eliminating redundancy, and supporting scalability.